

# 08 Messhilite Inheritance Analysis - Translation and Revision Report

## Scope

This report records the English reconstruction of Japanese Chapter 08 and its supplementary RF5 validation package. The English edition preserves the original 8-series topic while updating its claims to match the research boundaries established in Chapters 06 and 07.

## Sources checked

- Japanese Chapter 08 source PDF (30 pages)
- Validation Methodology PDF
- Wilson Confidence-Interval Derivation PDF
- Aggregated Results PDF
- Condition-level PDFs: 3-stack through 6-stack
- Integrated Validation CSV (4,000 trial records)
- Existing English overview: [articles/Messhilite-Inheritance.md](#)
- User-provided current article draft and current Research Review decisions for Chapters 06 and 07

## Principal retained observations

- RF5: 1,000 trials each for 3-stack, 4-stack, 5-stack, and 6-stack conditions.
- Success counts: 250 / 387 / 519 / 571.
- Observed rates: 25.0% / 38.7% / 51.9% / 57.1%.
- Wilson intervals retained from the validation package.
- Reference values: 25.0% / 40.0% / 50.0% / 57.1%.
- Same-material versus different-material distinction.
- RF4SP small-sample limitation.
- Love Crystal performance comparison: 9% versus 6%.
- RF5 upgrade-after-inheritance observation.
- Directional weapon-category / shield-performance observation: the performance-source category, rather than only the completed weapon's visible category, was consistent with the observed shield result.
- Reproducible data-collection method and possible future community collection.

## Research-boundary revisions

### Candidate Count Model

The model is presented as a combination-space model, not a universal success-probability law.  $C(T,3)/C(N,3)$  is explicitly labeled a simple equal-selection reference approximation.

### Equal selection

The RF5 Messhilite results are described as consistent with the reference model under the tested conditions. The chapter does not infer that all RF5 candidate-overflow cases are uniformly distributed.

### Subset and order

Candidate-subset success and final order are kept analytically distinct. The text does not claim that they are implemented as separate internal processing stages.

### RF4SP generalization

The five-trial RF4SP checks are retained only as preliminary directional evidence. The English edition does not assign them precise probability estimates or infer cross-title implementation identity.

## Object X and turnip-family observations

The Chapter 06 turnip-family subset bias and Chapter 07 Mealy Apple position bias are used as boundary evidence. They do not overwrite the RF5 Messhilite observations; they prevent universalization of the equal-selection approximation.

## Practical interpretation added

- 4-stack: efficiency-oriented recommendation because it is the first stack above the three target slots and improved the observed rate by 13.7 percentage points.
- 5-stack: stability-oriented recommendation because it reached 51.9% and retained a further 13.2-point improvement.
- 6-stack: highest tested rate, with diminishing marginal gain from 5-stack.

These recommendations are separated from research observations and limited to comparable same-material Messhilite conditions.

## Additional weapon-category observation

The English edition adds the current shield-performance observation as a separate performance boundary rather than mixing it into the probability model. Shield-compatible performance sources could express shield performance after inheritance into dual blades or fists, while the reverse direction did not restore the expected shield performance. The wording is limited to the tested Messhilite conditions and does not generalize the result to every category-dependent mechanic.

## Terminology

- stack identifies the number of copies of the same target material.
- total candidates includes Messhilite Ore and is therefore stack count + 1 in the primary series.
- successful subset means a retained three-candidate subset satisfying the declared target condition.
- reference probability replaces language that could imply a proven internal lottery rule.
- Unknown is retained for internal candidate generation, weighting, selection, ordering, and transfer mechanisms.

## Numerical checks

- Raw integrated CSV contains 4,000 records, 1,000 per condition.
- Success totals match the published summaries.
- $C(3,3)/C(4,3) = 1/4 = 25.0\%$ .
- $C(4,3)/C(5,3) = 4/10 = 40.0\%$ .
- $C(5,3)/C(6,3) = 10/20 = 50.0\%$ .
- $C(6,3)/C(7,3) = 20/35 = 57.1\%$ .

## Status

- 08 English reconstruction: completed.
- Source and data reconciliation: completed.
- Mathematical boundary revision: completed.
- Local DOCX/PDF visual verification: PASS. Main document: A4, 6 pages. Translation Report: A4, 2 pages. Every rendered page was inspected after table reconstruction.
- Research Review: pending.